MIL-L-19844A
27 DECEMBER 1963

SUPERSEDING
MIL-L-19844(Aer)
4 FEBRUARY 1957
MIL-S-5458A
18 APRIL 1951

MILITARY SPECIFICATION

LEADS; SHIELDED, DETACHABLE IGNITION

This psecification has been approved by the Department of Defense and is Mandatory for use by the Departments of the Army the Navy, and the Air Force.

L SCOPE

1.1 This specification covers shielded detachable ignition leads for reciprocating engines.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NN-P-515	— Plywood, Container Grade
PPP-B-585	— Boxes, Wood, Wirebound
PPP-B-591	—Boxes, Fiberboard, Wood-Cleated
PPP-B-601	— Boxes, Wood, Cleated-Plywood
PPP-13-621	— Boxes, Wood, Nailed and Lock-Corner
PPP-13-686	— Box, Fiberboard
PPP-T-60	— Tape, Pressure Sensitive Adhesive, Waterproof for Packaging and Sealing

Miliatary

MIL-P-116	— Preservation, Methods of
MIL-C-3702	— Cable, Power, Electrical, Ignition, High-Tension
MIL-I-6181	— Interference Control Requirements, Aircraft Equipment
MIL-S-7742	— Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-L-10547	—Liners, Case, and Sheet, Overwrap, Water - Vaporproof or Waterproof, Flexible
MIL-D-70327	— Drawings, Engineering and Associated LIsts

FSC 2925

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STANDARDS

MILITARY

MIL-STD-129- Marking for Shipment and Storage

MIL-STD-130- Identification Marking of US Military Property

MS21267 — Terminal Assembly, Detachable Ignition Lead

MS25065 — Ferrule, Flexible Conduit, Radio Frequency Shielding

AIR FORCE-NAVY AERONAUTICAL

AN4060 — Protector - Ignition Lead Terminal

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Qualification. The shielded detachable ignition leads furnished under this specification shall be a product which has been subjected to and which has passed the qualification tests specified herein, and which has been listed on or approved for listing on the applicable Qualified Products List.

3.2 Materials, Materials shall be suitable for the purpose intended.

3.2.1 Metals. Except as specified on Standard MS21267, all metals used in the construction of shielded detachable ignition leads shall be of a corrosion-resistant type, Aluminum and aluminum alloys shall not be used. Soldered or brazed joints shall be suitable for an ambient temperature of 450° F.

3.2.2 Nonmetallic. Nonmetallic materials shall be nonhygroscopic and suitable for ambient temperatures ranging from —65° to +450° F.

3.3 Design and construction. The leads shall be constructed to withstand the normal strains of jars, vibrations, and other conditions incident to shipping, storage, installation, and service, without failure. The leads shall consist of flexible conduit, ferrules, nuts, cable and terminals as follows.

3.3.1 Conduit. The conduit shall consist of a watreproof, flexible sheath covered by a uniform, woven wire braid of one or more layers and having at least five strands in each carrier.

3.3.1.1 Splices. Splices in strands shall be neat and shall cause no sharp or rough projections on the conduit, There shall be no more than two splices per lead.

3,3.1.2 Brai id, The braid shall fit the sheath closely enough to prevent bunching or undue slipping yet permit bending of the lead to a 2-inch radius without damage.

3.3.2 Ferrules. Ferrules for spark plug terminals shall conform to Standard MS-21267. The shank of the ferrule may be straight or in the form of an elbow, as required. ferrules for the coil or manifold terminal shall conform to Standard MS-25065, except that the material shall be stainless steel.

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3.3.3 Nuts. Nuts shall conform to Standard MS21267.

3,3.4 Threads, All straight screw threads shall conform to Specification MIL-S-7742.

3.3.5 Terminals. Sleeves, springs, eyelets, retainers and studs for spark plug terminals shall conform to Standard MS21267. Terminals for coils and manifolds shall be designed to fit receptacles in these specific items. Unless otherwise specified the spark plug terminal assembly shall be MS21267-L

3.3.6 Cable. The cable shall conform to Specification MIL-C-3702 and shall be type 3702-5MS designated therein.

3.3.7 Clip sizing sleeve. If support clips are to be attached to the lead assembly, a suitable nonmetallic sizing sleeve shall be attached to the assembly to prevent abrasion of the braid by the clip when the assembly is subjected to vibration,

3.3.8 Protective covers. When specified by the procuring activity (see 6.2), the leads shall be provided with a suitable oil, gasoline, and weather resistant covering over the metallic braiding. The covering shall provide adequate protection against chafing and abrasion when the leads are subjected to vibration.

3.4 Interchangeability, All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manuf acturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.5 Identification of product. Each ignition lead shall be durably and legibly marked in accordance with Standard MIIA3TD-180 either on the flats of the coupling nut, on the elbow, or on the shielding cover.

3.6 Performance.

3.6.1 Electrical leakage, The electrical leakage resistance between conductor and' shielding shall be not less than 1,500 megohms.

3.6.2 Air leakage. The terminal seal shall be capable of withstanding a differential air pressure of 25 psi without leakage or structural damage when the lead is connected as specified in 4.5.3.

3.6.3 High temperature, There shall be no evidence of damage that might affect subsequent performance as a result of the test specified in 4.5.4.

3.6.4 Cold bending. There shall be no evidence of damage that might affect subsequent performance as a result of the test specified in 4.5,5,

3.6.5 Moisture and maximum voltage, Upon completion of the test specified in 4.5.6, the leads shall pass the test specified in 4.5.2,

3.6.6 Engine endurance. There shall be no evidence of malfunctioning or of lead breakdown either during or as a result of the test specified in 4$5.7,

3.6,7 Shielding properties. Leads shall satisfy the requirements of Specification MIL-1-6181.

3.6.8 Bending Leads shall satisfy the requirements of 4.5.9,

4. QUALITY ASSURANCE PROVISIONS

4.1 responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification

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where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of shielded detachable leads shall be classified as:

(a) Qualification tests (4.3)

(b) Quality conformance tests (4*4)

4.3 Qualification tests.

4.3.1 Sampling instruction. Qualification test samples shall consist of a sufficient number of leads to equip the engine for which they are designed and 12 additional leads representative of the various lengths used on this engine. Three of the additional leads shall be not less than 24 inches in length. Samples shall be identified as required and forwarded to the activity responsible for qualification as designated in the letter of authorization (see 6.3).

4.3.2 Tests. Qualification tests of leads shall consist of the following tests conducted in the order listed:

(a) All qualification test samples

Examination of product	(4.5.1)
Electrical leakage	(4.5.2)
Air leakage	(4.5.3)

(b) Three long test samples

High temperature	(4.5.4)
Cold bending	(4.5.5)
Air leakage	(4.5,3)
Moisture and maximum voltage	(4.5.6)
Bending	(4.5.9)

(c) Engine set of leads

Engine endurance	(4.5.7)
Electrical leakage	(4.5.2)
Air leakage	(4*5,3)
Shielding properties	(4.5.8)
Moisture and maximum voltage	(4,5.6)

4.4 Quality conformance.

4.4.1 Individual tests. Each lead submitted for acceptance under contract shall be subjected’ to:

(a) Examination of product (4.5.1)

(b) Electrical leakage (4,5.2)

4.5 Test methods. Suitable service type mating receptacles shall be used during the conduct of these tests.

4..5.1 Examination of product. Each lead submitted for acceptance shall be carefully examined for conformance to the requirements of this specification for which no test methods are specified,

4.5.2 Electrical leakage. Leads shall be subjected to a dc potential of 15kv applied between the conductor and the shielding, The leakage resistance shall be measured.

4.5.3 Air leakage. Each terminal connector shall be installed in a suitatble service type mating receptacle. With the assembly immersed in water, air pressure at 25 psi shall be maintained in the terminal well for a period of 5 minutes (see 3.6.2).

4.5.4 High temperature. The leads shall be placed in a circulating air oven capable of reaching a temperature of 450° ±4° F within a period of not less than 1 hour nor more than 2 hours. The duration of temperature aging shall be 125 hours, Throughout the first and last 5-hour periods of the 125 hours of aging, the leads shall be subjected to a 60-cycle ac potential of 12kv rms applied between the conductor and the shielding, The initial voltage application shall be made simultaneously with the start of the oven heating units.

4.5.5 Cold bending. Three leads at least 24 inches long shall be placed in a cold box and maintained at a temperature of —650 + 00, —50 F for 24 hours. Immediately following this period, the leads, without removal from the cold box, shall be wound on a 3-inch diameter mandrel, The leads shall then withstand, while still at the specified low temper-

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ature, a 60-cycle ac potential of 12kv rms applied between the conductor and the shielding for 2 hours without electrical failure.

4.5.6 Moisture and maximum voltage. The leads shall be bent to an inner radius of 2 inches and their ends restrained to maintain this configuration. The entire braided portion of the leads shall then be immersed in water for a period of 30 minutes during which a potential of 12kv rms at 60 cycles shall be applied between the conductor and the shielding. The leads shall then be subjected to the electrical leakage test, 4.5.2.

4.5.7 Engine endurance The set of leads for the engine test shall be installed on the engine for which they were designed and subjected to a 150-hour engine test. The leads shall be observed for evidence of malfunctioning or breakdown during this test. Upon completion of this test the leads shall be removed from the engine and examined for evidence of damage or wear. The leads shall then be tested and meet the requirements of 4.5.2 and 4.5.3.

4.5.8 Shielding properties. Leads shall be tested in accordance with Specification MIL 1–6181 and the shielding of the leads shall not permit radio interference in excess of the limits specified therein,

4.5.9 Bending. The elbow or straight fitting shall be rigidly attached to a fixture at the conduit end and a bending torque applied through a level securely coupled to the spark plug coupling nut and extending along the axis of this outlet. The lever arm is measured from the center of the elbow or straight fitting conduit joint. Torque shall be applied in both directions in the plane of the elbow, and at a right angle thereto. Elbows and straight fittings shall be capable of withstanding 175 pound-inches of torque in each direction without def ormation to the body of the fitting.

4,6 Packaging, packing, and marking. Preparation for delivery shall be examined for conformance to section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2). Prior to packaging, the terminal connectors of each lead shall be inserted in protectors conforming to Standard AN4060 of the part number specified (see 6.2),

5.1.1 Level A, Unless otherwise specified, five leads of the same size and part number shall be packaged in a unit package in accordance with Method IA of Specification MIL-P-116, without the use of a preservative compound, Six unit packages shall be further packaged within an intermediate container conforming to Specification PPP-B-636 and sealed in accordance with the appendix thereto,

5.1.2 Level C. Packaging of leads shall be such as to provide protection against deterioration or damage during handling and shipment from supply source to first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Leads, packaged as specified, shall be packed in overseas-type exterior containers conforming to Specification PPP-B-591, PPP-13-601, PPP-B-621, 01’ PPP-B-636 (type I or II, class 2). Plywood, when used, shall be type I or II, class 2 of Specificat ion NN-P-515. Strapping and closures shall be in accordance with the appendix to the applicable container specification, Containers conforming to Specification PPP-B-591 or PPP-B-621 shall be provided with a ease liner conforming to Specification MIL L-10547 and sealed in accordance with the appendix thereto. The case liner may be omitted when the unit or intermediate container conforms to class 2 of Specification PPP-B-636; provided the center and edge seams and the manuf acturer’s joints are sealed with tape at least 1½ inches wide,

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conforming to type III, class 1 of Specification PPP-T-60, The gross weight of wood-type shipping containers shall not exceed 200 pounds, When fiberboard containers are utilized, the fiberboard shall have a minimum Mullen test of 275 pounds and the gross weight shall not exceed the weight limitations of the applicable container specification.

5.2.2 Level B. Leads, packaged as specified, shall be packed in domestic-type exterior containers conforming to Specification PPP-B-585 (class 1), PPP-B-591 (style B), PPP-B-601 (style B), PPP-B-621 (class 1, style 2), PPP-B-636 (type I or II, class 1). Closures shall be in accordance with the appendix of the applicable container specification. When fiberboard containers are utilized, the fiberboard shall have a minimum Mullen test of 275 pounds, The gross weight of the shipping container, when packed for shipment, shall not exceed 200 pounds; the gross weight of fiberboard containers shall not exceed the weight limitations of the applicable container specification.

5.2.3 Level C, Leads shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest applicable rate. Containers shall be in accordance with Uniform Freight Classification Rules, or regulations of other carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Unit and intermediate packages. Each unit and intermediate package shall be legibly and durably marked with the following information in such manner that the markings will not become damaged when the packages are opened:

Stock number
LEADS ; SHIELDED, DETACHABLE IGNITION
Specification MIL-L-19844
Quantity
Length
Manufacturer's part number
Date of manufacture (month and year)
Name of manufacturer
Name of contractor (if different from manufacturer)
Contract or order number
For use on engine models (state applicable engine model numbers)

5.3.2 Shipping containers. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL STD-129.

6. NOTES

6.1 Intended use. Shielded detachable ignition leads are intended for use with aircraft engine ignition radio shielding assemblies as easily replaceable ignition leads.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.

(b) Specify MS21267-2, if desired (see 3.3.5) .

(c) Whether or not an outer protective cover is required (see 3.3.8).

(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(e) Engine model designation for which leads are intended,

(f) Engine cylinder number and whether front or rear lead.

(g) Specify part number of terminal protector desired (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that

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elate, The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they maybe eligible to be awarded contracts or orders for the products covered' by this specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Navy Department, Washington, D. C. 20390, and information pertaining to qualification of products may be obtained from that activity.

6.4 Marginal indicia. The margin of this specification are marked to indicate where changes deletions, or additions to the previous issue have been made, This is done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Figures are not so marked, Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army—MO
Navy-Weps
Air Force-AFLC (82)

Preparing activity:
Navy-Weps

Project No. 2925-0003

Reviewer activity:
Army—MO
Navy-Weps
Air Force-AFLC (82)

User activity:

Review/user information is current as of the date of this document, For future coordination of changes to this document, draft circulation should be based on the information in the current DoD Index of Specifications and Standards,

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-L-19644A Leads; Shielded, Detachable Ignition

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS OF ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD Form 1426

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